

Partial Least Squares (PLS) vs Principal Component Regression (PCR)

BESSANH Isaac

1 Introduction to Partial Least Squares (PLS) Regression

1.1 Brief History

The Partial Least Squares (PLS) method, initially proposed by Herman Wold in the 1960s, is a dimension reduction technique combining Principal Component Analysis (PCA) and regression. It was developed to estimate statistical relationships in contexts where classical linear regression fails, notably in the presence of multicollinearity or when the number of predictors greatly exceeds the number of observations. The method was further developed and popularized in the 1980s by Svante Wold, particularly in chemometrics, where data are often highly correlated and high-dimensional. Today, PLS is widely used across fields such as marketing, neuroscience, bioinformatics, energy, and economics for its robustness and predictive capabilities.

1.2 Why Use the PLS Method?

Consider a dataset with I observations, J predictors, and K dependent variables. Denote:

$$X \in \mathbb{R}^{I \times J} \quad \text{and} \quad Y \in \mathbb{R}^{I \times K},$$

with the primary objective being to regress Y on X . Achieving this requires considering several scenarios. When the matrix X is of full rank, i.e., its columns are linearly independent and Y is a vector, a multiple regression suffices. However, in many real-world applications, explanatory variables are highly correlated, meaning $J > I$, causing $X^\top X$ to be singular, and ordinary least squares estimation becomes impossible or unstable. To address this, several methods can be used, such as stepwise regression (forward or backward) or PCR. Principal Component Regression (PCR) addresses this issue by constructing principal components that maximize the variance of X . However, these components are determined in an unsupervised manner, i.e., without knowledge of Y (see section 6.3.2 of "An Introduction to Statistical Learning"). As a result, PCR may select directions that explain X well but are less informative for the relationship between X and Y , leading to the development of PLS to overcome this limitation.

PLS, in contrast, determines predictors of Y in a supervised manner by extracting latent components that summarize the useful information in X while maximizing the covariance between X and Y . In other words, PLS components are specifically constructed to optimally explain the predictive structure linking the two matrices.

1.3 Mathematical Formulation and PLS Algorithm

PLS regression assumes that the data $X \in \mathbb{R}^{n \times p}$ (centered, optionally standardized) and the response $y \in \mathbb{R}^n$ can be represented using a reduced number of latent components:

$$X = ZP^\top + E, \quad y = Zd + e,$$

where:

- $Z = [z_1, \dots, z_r]$ contains the PLS components,

- $P = [p_1, \dots, p_m]$ is the loading matrix for X ,
- $d = (d_1, \dots, d_m)^\top$ contains the coefficients linking each component to y ,
- E and e are residuals on X and y ,
- r is the total number of principal components and $r = \text{rank}(X)$.

2 PLS Algorithm (Helland Version, Detailed)

The PLS algorithm is implemented following the steps described above.

Assume first that the variables X and y are centered before initializing:

$$X_0 = X, \quad y_0 = y.$$

For $h = 1, \dots, r$ (where $r = \text{rank}(X)$):

1. Weight Vector Calculation

$$w_h = X_{h-1}^\top y_{h-1}.$$

We calculate the covariance between each variable in X and the residual response y_{h-1} . This is the direction that maximizes covariance with y_{h-1} , normalized as

$$\|w_h\| = 1$$

to prevent numerical instability.

2. PLS Component

$$z_h = X_{h-1} w_h.$$

Each observation is projected onto the direction w_h . z_h is the h -th PLS component.

3. X Loadings

$$p_h = \frac{X_{h-1}^\top z_h}{z_h^\top z_h}.$$

Each variable in X_{h-1} is regressed on the component z_h . The vector p_h indicates how the component reconstructs X .

4. Regression Coefficient of y on z_h

$$d_h = \frac{y_{h-1}^\top z_h}{z_h^\top z_h}.$$

This is the coefficient from the simple regression of y on z_h .

5. Deflation of X

$$X_h = X_{h-1} - z_h p_h^\top.$$

We remove from X_{h-1} the part explained by the component z_h .

6. Deflation of y

$$y_h = y_{h-1} - d_h z_h.$$

We remove the part of y explained by the new component.

At the end, all components can be expressed as linear combinations of the original variables:

$$z_h = Xw_h,$$

and the component matrix is:

$$Z = XW.$$

The final model is thus:

$$\hat{y} = Zd = X(Wd).$$

The strength of PLS lies in constructing latent components optimized for predicting Y , making it robust and effective in high-dimensional or highly collinear settings.

2.1 Theoretical Comparison Between PLS and PCR

In statistical modeling, classical linear regression often becomes unstable when the set of explanatory variables (X) is large and highly collinear. Two commonly used dimension reduction methods combined with regression are Principal Component Regression (PCR) and Partial Least Squares Regression (PLS). Although both rely on constructing new variables, called latent components, they differ in their objectives and extraction mechanisms.

PCR clearly separates dimension reduction from regression. It begins by applying Principal Component Analysis (PCA) solely to the predictor matrix X and uses the resulting components to explain y . PCR's fundamental goal is to maximize the variance of predictors X . However, PCR completely ignores the response variable Y during latent component construction.

PLS, in contrast, directly links dimension reduction to the response variable Y . Its primary objective is to maximize the covariance between predictors X and the response Y .

This difference leads to a key distinction: PCR aims to best explain the internal structure of X but does not guarantee predictive relevance for Y . PLS, however, is explicitly optimized for predicting Y , as each component is constructed to maximize its relationship with the response variable. Consequently, in situations with many highly correlated variables (multicollinearity), PLS is generally considered the more suitable tool when the main goal is prediction.

2.2 Practical Example of PLS

2.2.1 Data Description and Problem Statement

We used a simulated dataset, containing **100 explanatory variables** observed over **80 observations**. We remark that the number of variable is larger than number of observation. So the OLS model is no longer appropriate for estimation. We will therefore move on to information reduction using PLS or PCR and then proceed to estimate both models in order to choose the best one.

2.2.2 PLS Model Performance and Interpretation

$$\mathbf{y} = \mathbf{X}\beta_{PLS} + \epsilon \quad (1)$$

Where:

β_{PLS} is the vector of regression coefficients in the original variable space, ϵ is the error term.

$$\hat{y}_{PLS} = -0.1185 Z_2 + 0.2907 Z_3 + 0.2232 Z_4 - 0.0934 Z_5 + \dots - 0.0460 Z_{99} - 0.3583 Z_{100} \quad (2)$$

The Partial Least Squares (PLS) model demonstrated exceptional performance on the training dataset, as evidenced by the following key metrics:

- **R² (Coefficient of Determination):** $R^2_{PLS} = 1.0$

- **MSE (Mean Squared Error):** $\text{MSE}_{\text{PLS}} = 6.097840678726984 \times 10^{-28}$

An R^2 value of 1.0 indicates that the PLS model explains 100% of the variance in the dependent variable (Y) using the independent variables (X). Coupled with an extremely low Mean Squared Error (MSE) that is virtually zero, this suggests a near-perfect fit of the model to the training data. In essence, the PLS model was able to reproduce the target variable values with remarkable precision on the dataset it was trained on.

Such a result highlights the strength of PLS in scenarios characterized by high dimensionality and multicollinearity, where it is specifically designed to extract latent components that are highly relevant for prediction. The model effectively captures the underlying relationships within the training data, demonstrating its robustness and predictive power in this context.

2.2.3 Comparison Between PLS and PCR

This section presents a comparative analysis of the Partial Least Squares (PLS) and Principal Component Regression (PCR) models, based on their performance metrics (R^2 and MSE) obtained from the training dataset.

$$\hat{y}_{\text{PCR}} = 1.26 \times 10^{12} Z_2 - 4.20 \times 10^{12} Z_3 + 3.91 \times 10^{12} Z_4 - 2.88 \times 10^{12} Z_5 + \dots + 9.17 \times 10^{13} Z_{99} - 7.92 \times 10^{12} Z_{100} \quad (3)$$

- **R^2 (Coefficient of Determination):** $R^2_{\text{PCR}} = 0.976483882884597$
- **MSE (Mean Squared Error):** $\text{MSE}_{\text{PCR}} = 61.585045738388054$

From the summarized metrics, it is evident that the **PLS model significantly outperformed the PCR model** on this specific training dataset. The PLS model achieved a near-perfect fit, as indicated by an R^2 value of 1.0 and an extremely low MSE of approximately 6.1×10^{-28} . This means the PLS model was able to explain virtually all the variance in the dependent variable.

In contrast, the PCR model, while demonstrating a strong performance with an R^2 of 0.976, showed a considerably higher MSE of 61.585. This implies that although PCR explains a large proportion of the variance, its predictive accuracy on the training data is not as precise as that of PLS.

2.3 Conclusion

For this project, where the objective is predictive modeling in a high-dimensional context, the PLS model proved to be significantly more effective than the PCR model. Its explicit focus on maximizing the covariance with the response variable allowed it to achieve a substantially better fit to the training data. This highlights PLS as a powerful tool for regression problems, particularly when dealing with complex datasets characterized by numerous, potentially collinear, predictors where predictive accuracy is paramount.